ATTENDANCE ACTIVITY (INFERENTIAL REVIEW QUESTION)

- I invented a new medication called "MoodX," which I hope will increase people's happiness levels. To test this, I will be randomly assigning 100 participants to either my MoodX group, or a group that takes a placebo (fake) pill. Then, I will compare these groups' scores of happiness (on a scale of 1-5) to see if my MoodX group showed higher levels of happiness than the placebo group.
- Using the example above, identify the following:
  1. What is the independent variable (IV) in this study? What is the dependent variable (DV)?
  2. What type of research design is this? (e.g., Between-subjects or Within-groups?)
  3. What is the Null hypothesis (H_0)?
  4. What is the Alternative hypothesis (H_A)?
  5. Is it one-tailed or two-tailed hypothesis?
  6. What would be an example of a Type I Error?
  7. What would be an example of a Type 2 Error?
  8. Let's imagine I found that the MoodX group had significantly higher scores of happiness compared to the control group!! Would this be considered "rejecting the null hypothesis" or "failing to reject the null hypothesis?" Why?

LET'S REVIEW ☺

1. What is the independent variable (IV) in this study? What is the dependent variable (DV)?
   – IV: Pill (MoodX or placebo)
   – DV: Happiness level (score of 1-5)
2. What type of research design is this? (e.g., Between-subjects or Within-groups?)
   – Between-subjects design
3. What is the Null hypothesis (H_0)?
   – There will be *no difference* in happiness scores between the MoodX group and the placebo group.
4. What is the Alternative hypothesis (H_A)?
   – The MoodX group will show *statistically significantly higher* scores of happiness than the placebo group.
5. Is it one-tailed or two-tailed hypothesis?
   – One-tailed hypothesis

LET'S REVIEW ☺

6. What would be an example of a Type I Error?
   - It would be if I said MoodX DOES made people happier, when it actually does NOT.

7. What would be an example of a Type 2 Error?
   - It would be if I said MoodX does NOT make people happier, when it actually DOES.

8. Let's imagine I found that the MoodX group had significantly higher scores of happiness compared to the control group!! Would this be considered "rejecting the null hypothesis" or "failing to reject the null hypothesis?" Why?
   - Rejecting the null hypothesis

STATISTICAL DESIGNS

T-TESTS AND ANOVAS

KNOW THE DIFFERNECE

- STATISTICAL DESIGN
- EXPERIMENTAL DESIGN

STATISTICAL DESIGNS

- t-Test: used to compare the means of 2 groups and how they are related
  - One sample
  - Paired/dependent
  - Independent
- ANOVA: used to compare the means of 2+ groups
  - One Way
    - Between Subjects
    - Within-subjects

T-TESTS USE

- Used to compare means of 2 groups

T- TEST: FIRST TYPE

- One Sample T-Test: used to determine if the mean of a single sample significantly differs from a known population mean

- Comparison
  - Known
  - Logical

ONE SAMPLE T-TEST EXAMPLE

PARTICIPANT	HEART RATE
1	60
2	58
3	49
4	70
5	55

DO LONG DISTANCE RUNNERS HAVE LOWER RESTING HEART RATES?

We know that average resting heart rate = 75

INDEPENDENT SAMPLES T- TEST

- Used to determine if there is a significant difference between the means of two independent groups
- Two means
- Random assignment or class variables
- Between-subject variables

EXAMPLE INDEPENDENT-SAMPLES T- TEST

- Is there a difference between men and women resting heart rate?

MEN		WOMEN	
Subject #	Score	Subject #	Score
1	75	11	56
2	58	12	82
3	59	13	60
4	53	14	53
5	82	15	72
6	84	16	72
7	65	17	64
8	83	18	73
9	82	19	82
10	81	20	82

PAIRED SAMPLE T-TEST

- Used to compare the means of two related samples, where the observations are paired or dependent

- Two datapoints
  - NOT independent
    - Same people measured before and after something
    - Same people measured in two conditions

EXAMPLE PAIRED SAMPLES T-TEST

- Predict a difference in resting heart rate based on emotional content

Participant	Standard	Emotional
1	64	72
2	73	83
3	65	62
4	83	93
5	72	82
6	64	72
7	55	64
8	73	73
9	72	82
10	81	82

ANALYSIS OF VARIANCE (ANOVA)

- Use if more that 2 groups

EXAMPLE-ANOVA

- What type of program has the greatest impact on aggression?
    - Violent movies, soap operas, or “infomercials”?
    - *IV?*
    - *DV?*

BETWEEN-SUBJECTS ANOVA: EACH SCORE IS INDEPENDENT

Participant	Soap Opera	Participant	Violent Movies	Participant	Infomercial
1	10	5	15	9	20
2	15	6	25	10	17
3	20	7	30	11	25
4	13	8	25	12	20

WITHIN-SUBJECTS ANOVA: SAME PEOPLE IN MULTIPLE CONDITIONS

Participant	Soap opera	Violent movies	Infomercial
1	10	15	20
2	15	25	17
3	20	30	25
4	13	25	20

ONE-WAY ANOVA: BETWEEN-SUBJECTS VS. WITHIN-SUBJECTS

- Both are used when:
  - We compare 3 or more means
  - There is one independent variable
- Between-subjects ANOVA
  - Different people are in each group
  - Each participant gives one score
- Within-subjects ANOVA
  - The same people complete all conditions
  - Each participant gives multiple scores

QUICK REVIEW (ATTENDANCE ACTIVITIES)

ATTENDANCE ACTIVITY (T-TEST & ANOVA REVIEW #1)

- Dr. Blevins wants to know if her introductory statistics class has an above average understanding of basic math. 15 students are chosen at random from the class and given a math proficiency test. Dr. Blevins knows that an average student's understanding would be scoring a 75% on the test.
- What statistical design would Dr. Blevins conduct to determine if her class has an above average understanding of basic math?

ATTENDANCE ACTIVITY (T-TEST & ANOVA REVIEW #2)

- Cleo predicts that students will learn most effectively with constant background sound, as opposed to an unpredictable sound or no sound at all. She randomly divides 24 students into 3 groups (Groups A, B, and C). All students study a passage of text for 30 minutes. Those in Group A study with background noise being played at a constant volume. Those in Group B study with noise that changes volume periodically. Those in Group C study with no noise at all. After studying, all students take a 10 point multiple choice trivia test.
- Which statistical design should Cleo use to test her hypothesis?

ATTENDANCE ACTIVITY (T-TEST & ANOVA REVIEW #3)

- Dr. Phiat theorizes that people who participate in a regular program of exercise will have levels of blood pressure that are significantly different from that of people who do not participate in a regular program of exercise. To test this idea. Dr. Phiat randomly assigns 21 participants to an exercise program for 10 weeks, and 21 participants to a non-exercise comparison group. After 10 weeks, the mean blood pressure of participants will be measured.
- What statistical design should Dr. Phiat use?